MATH329 · OPERATIONS RESEARCH

Lecture 1 Introduction to OR

Midterm summary · Chapter 1

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01 What is Operations Research?

The course has no derivatives and no integrals. It uses simple linear equations to find the best decision: the highest profit or the lowest cost.

Operations Research (OR) is an interdisciplinary branch of applied mathematics and formal science that uses methods like mathematical modeling, statistics, and algorithms to arrive at optimal or near optimal solutions to complex problems.

Fact	What to remember
Where and when did it start?	In **England** during **World War II**, when a team of British scientists set out to decide the best use of war material.
What is it today?	A dominant and indispensable **decision-making tool**.
Is the mathematical model enough?	No. Unquantifiable **intangible factors**, such as human behavior, must be accounted for in the final decision.

Appeared in the midterm

Midterm 461 gave the sentence as a blank and the answer was **England**. Midterm 472 gave it as true/false with **World War I**, so the answer was False because it was the second war.

02 The parts of any model

Every model in the course has three parts, and this is the order you write them in:

#	Component	Meaning
1	**Decision variables**	The quantities we want to find, such as x_1 and x_2.
2	**Objective function**	The goal we maximize or minimize.
3	**Constraints**	The restrictions the solution must satisfy.

Term	Meaning
Feasible solution	Any values of the variables that satisfy all the constraints.
Optimum feasible solution	The feasible solution that gives the best value of the objective.

Appeared in the midterm

A true/false item that appeared in both 471 and 472:

> The solution of the LPP may be compatible with its constraints imposed by the available resources. (False: the solution **must** satisfy the constraints.)

03 Example 1.1: the Bader company

Bader manufactures men's shirts and women's blouses. It employs 25 workers in cutting, 35 in sewing and 5 in packaging. The factory works one 8-hour shift, 5 days a week. Determine the optimal weekly production schedule.

Garment	Cutting (min)	Sewing (min)	Packaging (min)	Unit profit ($)
Shirts	20	70	12	8
Blouses	60	60	4	12

1 Decision variables

x_1 = number of shirts per week

x_2 = number of blouses per week

2 Objective function

$$\text{Maximize } z = 8x_1 + 12x_2$$

3 Constraints

Available time per department in minutes = workers × 40 hours × 60 minutes. The 40 comes from 8 hours × 5 days.

$$\begin{aligned}
&\text{Cutting:} && 20x_1 + 60x_2 \le 25 \times 40 \times 60 = 60000 \\
&\text{Sewing:} && 70x_1 + 60x_2 \le 35 \times 40 \times 60 = 84000 \\
&\text{Packaging:} && 12x_1 + 4x_2 \le 5 \times 40 \times 60 = 12000 \\
&x_1, x_2 \ge 0, \text{ integers}
\end{aligned}$$

04 Linear Programming Problem (LPP)

An optimization problem is called a Linear Programming Problem (LPP) when the objective function and all the constraints are linear functions of the decision variables $x_1, x_2, \ldots, x_n$.

We also include the **non-negativity restrictions**, namely $x_j \geq 0$ for all variables. The general form:

Optimize (Maximize or Minimize)	$z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$		
subject to	$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n$	$(\leq, =, \geq)$	b_1
	$\ldots$		
	$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n$	$(\leq, =, \geq)$	b_m
	$x_1, x_2, \ldots, x_n \geq 0$		

The two properties of an LPP

Property	Meaning
Proportionality	The contribution of each variable to the objective and to the constraints is directly proportional to its value. If x_1 doubles, its contribution doubles.
Additivity	The total is the direct sum of the individual contributions of the variables.

Appeared in the midterm

Midterm 461 gave the definition as a blank and the answer was **LPP**. Midterm 472 changed one word in a true/false item:

> In a linear programming problem, the objective function **or** the constraints must be linear. (False: the objective function **and** all the constraints.)

05 The phases of an OR study

Five phases in this order. They can be asked as "list" or "order":

1. Definition of the problem
2. Construction of the model
3. Solution of the model
4. Validation of the model
5. Implementation of the solution

06 Formulation examples

Example 1.2 · Wild West hats

Wild West produces two types of cowboy hats. Type I hat requires twice as much labor as Type II. If all the available labor time is dedicated to Type II alone, the company can produce a total of 400 Type II hats a day. The market limits are 150 and 200 hats per day for the two types. The profit is $8 per Type I hat and $5 per Type II hat. Formulate the problem as an LPP so as to maximize the profit.

The hard part is the labor constraint. Let a Type II hat take c minutes, so a Type I hat takes $2c$, and the total available time is $400c$:

$$(2x_1 + x_2)\, c \le 400\, c \quad \rightarrow \quad 2x_1 + x_2 \le 400$$

Maximize $z = 8x_1 + 5x_2$
subject to $2x_1 + x_2 \leq 400$
$x_1 \leq 150$
$x_2 \leq 200$
$x_1, x_2 \geq 0$, integers

Feed Mix · a minimization problem

Each cow should get at least 15, 20 and 24 units of nutrients A, B and C. In feed of variety 1 (variety 2) the contents of A, B and C are 1 (3), 2 (2), 3 (2) units per kg. The costs are 2 Rs. and 3 Rs. per kg. How much of each variety should be purchased so that the expenditure is least?

The words **at least** mean $\geq$, and **least expenditure** means minimize:

Minimize $z = 2x_1 + 3x_2$		
subject to	$x_1 + 3x_2 \geq 15$	(nutrient A)
	$2x_1 + 2x_2 \geq 20$	(nutrient B)
	$3x_1 + 2x_2 \geq 24$	(nutrient C)
	$x_1, x_2 \geq 0$	

Trim Loss

A company has to manufacture the circular tops of cans. Two sizes, one of diameter 10 cm and the other of diameter 20 cm, are required. They are cut from metal sheets of dimensions 20 cm by 50 cm. The requirement of the smaller size is 20,000 and of the larger size is 15,000. Formulate as an LPP so that the number of sheets used is a minimum.

A sheet can be cut in three patterns, and each variable is the number of sheets cut with one pattern:

Pattern	10 cm tops	20 cm tops	Variable
1	10	0	x_1
2	2	2	x_2
3	6	1	x_3

$$\text{Minimize } z = x_1 + x_2 + x_3$$

$$\text{subject to} \quad 10x_1 + 2x_2 + 6x_3 \geq 20000$$

$$2x_2 + x_3 \geq 15000$$

$$x_1, x_2, x_3 \geq 0, \text{ integers}$$

07 A fixed method for formulation

#	Step	How to spot it in the question
1	Define the variables	What the question asks for: how many to produce or to buy.
2	Write the objective	Profit means max. Cost or expenditure means min.
3	One constraint per resource	"At most", "available", "limit" mean $\leq$. "At least", "requirement" mean $\geq$.
4	Match the units	If times are in minutes and availability in hours, convert first.
5	Do not forget the last line	$x_1, x_2 \geq 0$

08 Common mistakes

Mistake	Correct
Writing World War I	**World War II**
Saying the objective or the constraints are linear	The objective and all the constraints are linear.
Deciding max or min from the inequality sign	It is decided only by the objective's wording: profit or cost.
Forgetting to convert hours to minutes	`25 × 40 × 60 = 60000`
Forgetting $x_1, x_2 \geq 0$	It is written at the end of every model.

09 Test yourself

1. Where and when did the formal activities of OR start?
2. Name the three components of any model.
3. When is a problem called an LPP?
4. What is the difference between proportionality and additivity?
5. A bakery makes cake and bread. A cake needs 2 oven hours and 1 kg of flour and earns 30. A loaf needs 1 oven hour and 2 kg of flour and earns 20. There are 10 oven hours and 8 kg of flour. Write the model.

Answers

#	Answer
1	In **England** during **World War II**.
2	**Decision variables**, **Objective function**, **Constraints**.
3	When the objective function and all the constraints are linear functions of the decision variables.
4	The first: a variable's contribution is proportional to its value. The second: the total is the sum of the contributions.
5	Max $z = 30x_1 + 20x_2$ $2x_1 + x_2 \leq 10$ $x_1 + 2x_2 \leq 8$ $x_1, x_2 \geq 0$